

SINGLE-LEG STANCE



Test Objective for the Single Leg Stance:

The Single-Leg Stance assessment evaluates the ability to stabilize independently on one leg in a static posture.

How to Perform the Single Leg Stance:

The patient assumes the starting position by standing erect with feet together, toes pointing forward and arms relaxed to the sides of the thighs. Have the patient flex their right knee so their right foot comes off the ground. The patient maintains full upright posture balancing only on their left leg for at least 10 seconds. Repeat with eyes closed for 10 seconds. Repeat the test process for the right leg single-leg stance.

Functional Criteria:

- Maintains balance for 10 seconds eyes open
- Maintains balance for 10 seconds eyes closed
- Maintains posture
- Maintains foot position
- Without excessive effort and/or motor control

Common Compensations:

- Pelvic deviation
- Flails arms
- Moves original foot position

Tips for Testing:

- We expect a 50% increase in sway with eyes closed single-leg balance.
- Was there pain?



Special Note on Vision:

Vision is never a handicap to balance – it always helps. Any vision is better than no vision. Even a patient with severe cataracts will perform better with eyes open than eyes closed. Most experts cannot think of a single pathology in which removing vision would enhance postural sway.

SINGLE-LEG STANCE BREAKOUTS



Single-Leg Stance Breakout Rationale:

The Single-Leg Stance Breakout is broken down into two separate flowcharts. One is for the Vestibular System and Core Stability and the other one is for dysfunction in the ankle. We begin by checking the vestibular system.

Vestibular Problem

We check the vestibular system using the Clinical Test for Sensory Interaction on Balance (CTSIB). The CTSIB assesses the influence of vestibular, somatosensory and visual inputs on postural control. If the CTSIB is dysfunctional with a static head posture, a potential Clinical Vestibular Dysfunction (or Static Vestibular Dysfunction) is affecting the patient's Single-Leg Stance. Document this finding but continue on and proceed to the Half-Kneeling Narrow Base Test.

If the CTSIB is normal for static head posture, we proceed to dynamic head movements. Checking all three planes of motion (rotation, side bend and flexion/extension), we challenge the vestibular system one more time. If the CTSIB is now dysfunctional with a dynamic head movement, a potential Sub-Clinical Vestibular Dysfunction (or Dynamic Vestibular Dysfunction) is affecting the patient's Single-Leg Stance. Document this finding but continue on and proceed to the Half-Kneeling Narrow Base Test.

If both CTSIB variations are functional, then the vestibular system is not contributing to the problem. We then proceed to the Half-Kneeling Narrow Base Test.

Spine, Hip and Core Stability Problems

Using our SFMA logic of removing body parts, we can challenge the hip and core/spine for stability/motor control by removing the lower limb using the Half-Kneeling Narrow Base Test. If the patient can maintain a stable tall upright posture in the half-kneeling position and there is no pain, this demonstrates adequate stability/motor control through the hips and spine. Proceed to the Ankle Flowchart.

If the Half-Kneeling Narrow Base Test is dysfunctional and/or painful, a stability dysfunction needs to be addressed. Continue to change the stability requirements by moving the patient to quadruped position for the Quadruped Diagonals Test. This allows us to separate a weight-bearing spine and/or core stability/motor control dysfunction from a weight-bearing hip and/or core stability/motor control dysfunction. If the Quadruped Diagonals Test is functional, we have ruled in a weight-bearing spine and/or core stability/motor control dysfunction since we removed the load on the spine.

If Quadruped Diagonals Test is dysfunctional, then we have ruled in a hip and/or core stability/motor control dysfunction. If Quadruped Diagonals is painful, note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

Note - if hip extension and/or shoulder flexion presented as a mobility dysfunction in another breakout, treat those first!

No matter the result, we must continue to look at what we had removed - the lower limb. Proceed to the Ankle Flowchart.

SINGLE-LEG STANCE BREAKOUTS



Ankle Flowchart

Ankle Dorsiflexion and Plantarflexion Problems

Since we removed the lower limb in the Vestibular/Core flowchart, we must go back and assess ankle dorsiflexion (knee extended), ankle plantarflexion and inversion/eversion.

To assess both active ankle dorsiflexion (knee extended) and plantarflexion, we use a tandem position set up position.

If Active Tandem Dorsiflexion – Knee Extended Test is functional and non-painful, consider ankle dorsiflexion with knee extended normal and proceed to the Active Tandem Plantarflexion Test. If Active Tandem Dorsiflexion – Knee Extended Test is dysfunctional and/or painful, proceed to the Passive Prone Dorsiflexion – Knee Extended Test to differentiate between a mobility or stability/motor control dysfunction.

If Passive Prone Dorsiflexion – Knee Extended Test is dysfunctional, there is a dorsiflexion mobility dysfunction. If the test is now functional and non-painful, there is an ankle dorsiflexion SMCD. If passive dorsiflexion is painful (FP or DP), note where the pain is and come back and examine the region in more detail after completing the entire SFMA. Regardless of the ankle dorsiflexion finding, proceed to assess ankle plantarflexion.

If the Active Tandem Plantarflexion Test is functional and non-painful, consider ankle plantarflexion normal and proceed to the Active Seated Ankle Inversion/Eversion Test. If the Active Tandem Plantarflexion Test is dysfunctional and/or painful, proceed to the Passive Prone Plantarflexion Test to differentiate between a mobility and stability/motor control dysfunction.

If the Passive Prone Plantarflexion Test is dysfunctional, there is an ankle plantarflexion mobility dysfunction. If the test is now functional and non-painful, there is an ankle plantarflexion SMCD. If passive plantarflexion is painful (FP or DP), note where the pain is and come back and examine the region in more detail after completing the entire SFMA. Regardless of the ankle plantarflexion finding, proceed to assess ankle inversion/eversion.

Ankle Inversion and Eversion Problems

If the Active Seated Inversion/Eversion Test is functional and non-painful, consider ankle inversion/eversion normal. If this test is dysfunctional and/or painful, proceed to the Passive Inversion/Eversion Test to differentiate between a mobility and stability/motor control dysfunction.

If the Passive Inversion/Eversion Test is dysfunctional, there is an inversion/eversion mobility dysfunction. If the test is now functional and non-painful, there is an inversion/eversion SMCD. If either inversion or eversion is painful (FP or DP), note where the pain is and come back and examine the region in more detail after completing the entire SFMA.

Proprioception Problem

The only way to get a true diagnosis of a proprioceptive dysfunction is to rule out all other causes of a dysfunctional single-leg stance. This includes vestibular, core and ankle contributions. If ankle dorsiflexion, plantarflexion and inversion/eversion are functional and non-painful, and all

SINGLE-LEG STANCE BREAKOUTS

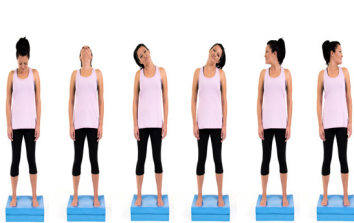


other assessments in the vestibular/core breakout are functional and non-painful, we can assume proprioceptive dysfunction is the primary problem. If there were previous findings in either of the two Single Leg Stance flowcharts, we cannot diagnosis as a proprioception dysfunction – we must treat those findings.

SINGLE-LEG STANCE BREAKOUTS



Selective Breakout Tests for Single-Leg Stance Breakout:



Vestibular Test - CTSIB



Half-Kneeling Narrow
Base Test



Quadruped Diagonals Test



Active Tandem
Dorsiflexion - Knee
Extended Test



Passive Prone
Dorsiflexion Test



Active Tandem
Plantarflexion Test



Passive Prone
Plantarflexion Test



Active Seated Ankle
Inversion Test



Passive Ankle
Inversion Test

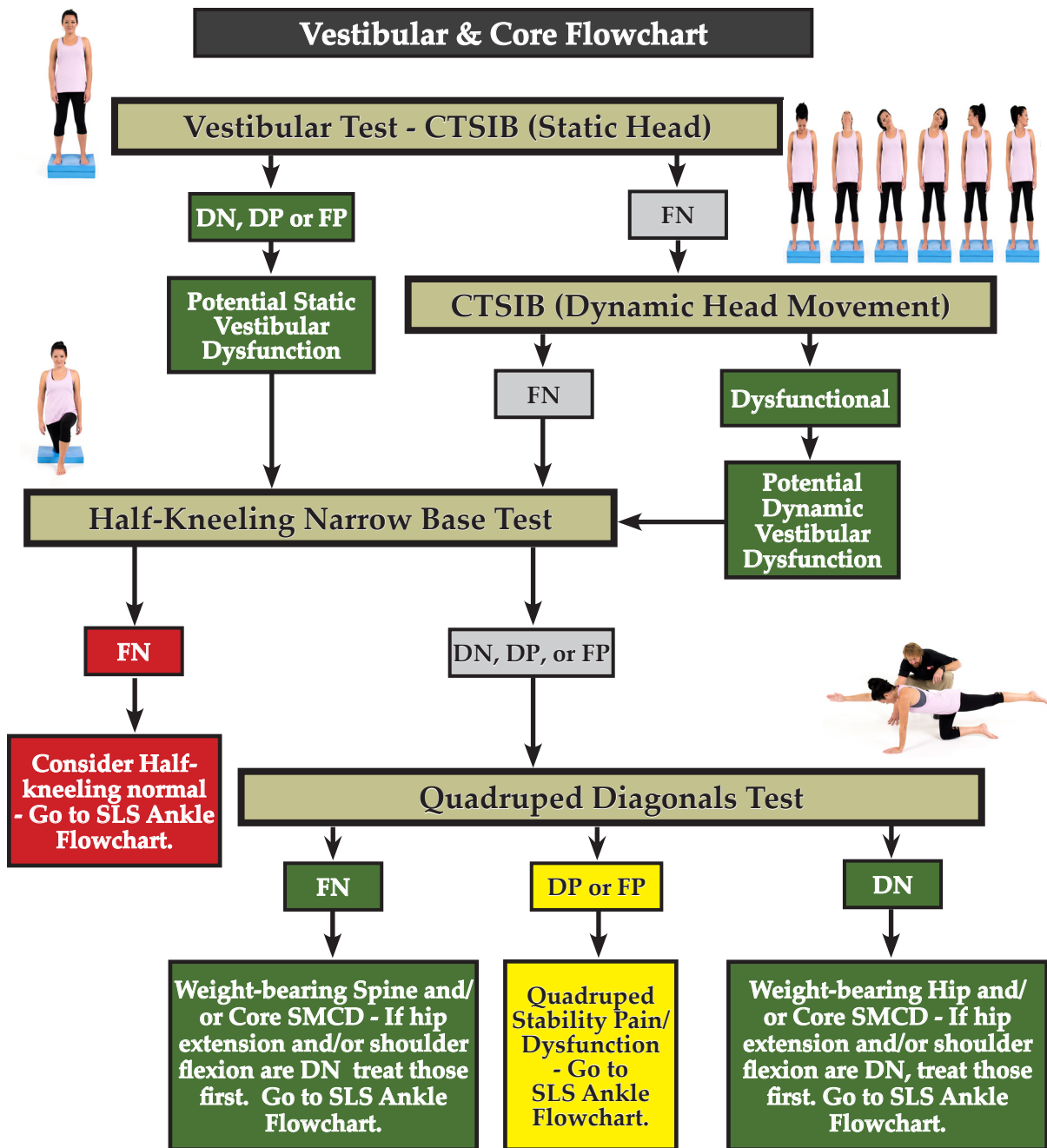


Active Seated Ankle
Eversion Test



Passive Ankle
Eversion Test

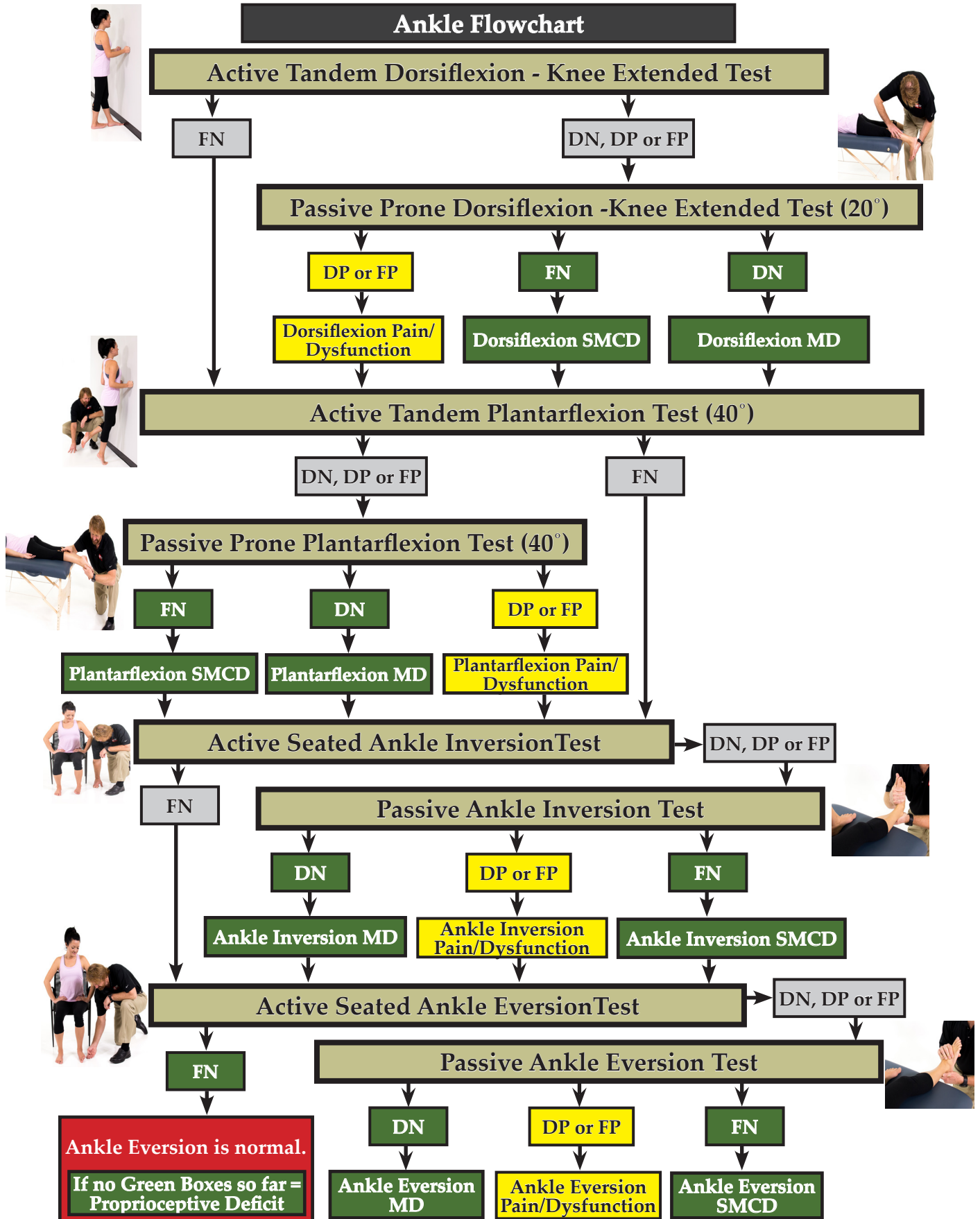
SINGLE-LEG STANCE BREAKOUTS



SINGLE-LEG STANCE BREAKOUTS



Ankle Flowchart



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the CTSIB (Static Head Movement) Test - Clinical Test for Sensory Interaction on Balance (CTSIB) Test:

This test is performed to rule out vestibular dysfunction. The Clinical Test for Sensory Interaction on Balance (CTSIB) is a complex sensory strategy balance test. The CTSIB assesses the influence of vestibular, somatosensory and visual inputs on postural control.

How to Perform the CTSIB (Static Head Movement) Test:

The CTSIB is administered by manipulating the support surface (firm versus foam) and visual conditions (eyes open versus eyes closed) while a patient is asked to maintain standing balance. The CTSIB assesses the patient's ability to maintain upright posture under four progressively more difficult sensory conditions. Each of these trials is timed for 20 seconds as the examiner watches for excessive postural sway or loss of balance.

- Condition 1: Normal base of support on a firm surface with eyes open.
- Condition 2: Normal base of support on a firm surface with eyes closed.
- Condition 3: Normal base of support on a foam surface (2 Airex Pads) with eyes open.
- Condition 4: Normal base of support on a foam surface (2 Airex Pads) with eyes closed.

Condition 1 provides a baseline reference with which the other three situation-dependent conditions will be compared.

Condition 2 removes vision, which provides information about the patient's ability to use somatosensory input to maintain upright posture.

Condition 3 disadvantages somatosensory input, which indicates the ability to use visual inputs to maintain upright posture.

Condition 4 disadvantages somatosensory input and removes vision, which indicates the ability to use vestibular input for maintaining upright posture.

If the patient passes Condition 4 of the CTSIB, the clinical sensitivity of the CTSIB can be enhanced by modifying the protocol to include dynamic head tilts. Requiring active rotation, flexion/extension and side bend head movements during upright stance generates visual and vestibular stimulation in addition to the sensory input created by postural sway. Altering the multisensory information increases the postural demands associated with maintaining upright stance. Consequently, the inclusion of head tilts can provide a more challenging balance task that could amplify subtle balance deficits.



SINGLE-LEG STANCE BREAKOUTS



How to Perform the CTSIB (Dynamic Head Movement):

Have the patient stand with a normal base of support on a foam surface (2 Airex Pads) with eyes open. Instruct the patient to:

- Tilt the head left and right. Repeat up to 5 times and stop.
- Tilt the head forward and backward. Repeat up to 5 times and stop.
- Rotate the head right and left. Repeat up to 5 times and stop.
- Repeat all head movements with eyes closed.



The complete sequence of head tilting should take less than 20 seconds. A metronome set at 120 beats per minute may be used to pace the patient.

Additional Information:

Many times, balance dysfunction can be due to vestibular system imbalances. This includes the inner ear's semicircular canals, which help indicate rotational movements, and the otoliths, which help indicate linear translations.

The vestibular system sends signals primarily to the neural structures that control eye movements, and to the muscles that keep us upright. The projections to the former provide the anatomical basis of the vestibulo-ocular reflex, which is required for clear vision, and the projections to the muscles that control our posture are necessary to keep us upright.

* Note - The CTSIB and modification with head tilting screens vestibular function as it relates to postural control. These tests do not screen for impairment in gaze stabilization, the vestibulo-ocular reflex.

Possible Findings:

- FN: An ability to stabilize with eyes open and eyes closed on the unstable surface with all three head movements, with no pain
 - » Indicates a normal functioning vestibular system. Proceed to Half-kneeling Narrow Base Test.
- DN: An Inability to stabilize with eyes open and eyes closed on the unstable surface with all three head movements, with/without pain
 - » Indicates a potential dynamic vestibular dysfunction and should be diagnosed by an appropriate clinician. Proceed to the Half-Kneeling Narrow Base Test.

*If pain is present with testing, use SFMA results to address pain and then retest.

SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Half-Kneeling Narrow Base Test:

To test the single-leg stance when mobility and stability requirements are reduced.

How to Perform the Half-Kneeling Narrow Base Test:

Instruct the patient to get into a half-kneeling position such that both the feet and the knees are in a straight line and the back foot is neutral. Repeat switching leg positions.

Functional Criteria:

The patient has 3 seconds to gain balance and then must hold the position solidly for 3 seconds while maintaining tall posture.

Additional Information:

An Airex pad can be used for comfort. The Airex also allows the ankle to remain in a neutral position.

Make sure the patient is not using their back foot for balance. Also monitor for shifting the body forward or back for stability.

Possible Findings:

- FN: An ability to maintain balance, no pain
 - » Consider stability/motor control of the hip and spine normal. Proceed to the SLS Ankle Flowchart.
- FP: An ability to maintain balance, pain
 - » Proceed to Quadruped Diagonals test.
- DN: An inability to maintain balance, no pain
 - » Proceed to Quadruped Diagonals test.
- DP: An inability to maintain balance, pain
 - » Proceed to Quadruped Diagonals test.



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Quadruped Diagonals Test:

To separate weight-bearing spine from a weight-bearing hip or core stability/motor control dysfunction.

How to Perform the Quadruped Diagonals Test:

Instruct the patient to get into a quadruped position such that the arms and thighs are at 90 degree angles to the torso - hands directly under shoulders, knees directly under hips. Have the patient reach out with their right arm and their left leg at the same time, maintaining balance with the left arm and right leg for a three second count. Repeat on the other side.

Functional Criteria:

Normal motion is maintaining balance for 3 seconds on each side, without rotation, excessive spine extension, or excessive effort.

Additional Information:

Limited hip extension or shoulder flexion could influence this test - we are not looking for full range. They should reach out as much as they can in order to evaluate balance.

Possible Findings:

- FN: An ability to perform the movement and maintain balance for 3 seconds, no pain
 - » Indicates a weight-bearing spine and/or core stability/motor control dysfunction. If hip extension/shoulder flexion is DN, examine it first. Proceed to the SLS Ankle flowchart.
- FP: An ability to perform the movement and maintain balance for 3 seconds, pain
 - » Indicates a painful weight-bearing spine and/or core stability/motor control dysfunction- complete local biomechanical exam. Proceed to the SLS Ankle flowchart.
- DN: An inability to perform the movement and maintain balance for 3 seconds, no pain
 - » Indicates a weight-bearing hip and/or core stability/motor control dysfunction. If hip extension/shoulder flexion are DN, examine it first. Proceed to the SLS Ankle flowchart.
- DP: An inability to perform the movement and maintain balance for 3 seconds, pain
 - » Indicates a painful hip and/or core stability/motor control dysfunction- complete local biomechanical exam. If hip extension/shoulder flexion are DN, examine it first. Proceed to the SLS Ankle flowchart.



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Active Tandem Dorsiflexion - Knee Extended Test:

To test dorsiflexion of the ankles in the sagittal plane with the knee in extension.

How to Perform the Active Tandem Dorsiflexion - Knee Extended Test:

Instruct the patient to assume the tandem gait stance using the wall for support. Tandem stance set-up is the left foot toes against the wall and the right foot toes against left heel, the 2nd/3rd toe in line with left heel. The elbows should be at 90 degrees and in line with the rib cage. Once in position, dorsiflex the right ankle. Switch legs for set-up and then repeat with left ankle.

Functional Criteria:

Normal motion is the base of the 5th metatarsal clearing the ground when dorsiflexing.

Additional Information:

They should be able to dorsiflex enough to clear the ground without excessive toe extension.

Watch for hyperextension of the knee - if present, cue to keep knee straight.

Possible Findings:

- FN: An ability to dorsiflex the ankle in tandem stance, no pain
 - » Consider ankle dorsiflexion normal. Proceed to the Active Tandem Plantarflexion.
- FP: An ability to dorsiflex the ankle in tandem stance, pain
 - » Proceed to the Passive Prone Dorsiflexion Test - Knee Extended.
- DN: An inability to dorsiflex the ankle in tandem stance, no pain
 - » Proceed to the Passive Prone Dorsiflexion Test - Knee Extended.
- DP: An inability to dorsiflex the ankle in tandem stance, pain
 - » Proceed to the Passive Prone Dorsiflexion Test - Knee Extended.

Starting foot position:



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Passive Prone Dorsiflexion - Knee Extended Test:

To test passive ankle dorsiflexion with knee extended in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Dorsiflexion - Knee Extended Test:

Instruct the patient to lie prone with the knee extended. Measure the total passive dorsiflexion of the ankle.

Functional Criteria:

Normal open chain ankle dorsiflexion with knee extended is 20 degrees.

Additional Information:

Before taking the measurement, make sure the ankle starts in a neutral position.

Possible Findings:

- FN: 20 degrees of passive ankle dorsiflexion with knee extended, no pain
 - » Indicates an ankle dorsiflexion stability/motor control dysfunction with knee extended. Proceed to the Active Tandem Ankle Plantarflexion Test.
- FP: 20 degrees of passive ankle dorsiflexion with knee extended, pain
 - » Indicates pain with ankle dorsiflexion with knee extended - perform local biomechanical exam. Proceed to the Active Tandem Ankle Plantarflexion Test.
- DN: Less than 20 degrees of passive ankle dorsiflexion with knee extended, no pain
 - » Indicates an ankle dorsiflexion mobility dysfunction with knee extended- complete local biomechanical exam. Proceed to the Active Tandem Ankle Plantarflexion Test.
- DP: Less than 20 degrees of passive ankle dorsiflexion with knee extended, pain
 - » Indicates a painful ankle dorsiflexion mobility dysfunction with knee extended - complete local biomechanical exam. Proceed to Active Tandem Ankle Plantarflexion Test.



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Active Tandem Ankle Plantarflexion Test:

To test plantarflexion of the ankles in the sagittal plane.

How to Perform the Active Tandem Ankle Plantarflexion Test:

Instruct the patient to assume the tandem gait stance using the wall for support. Tandem stance set-up is the left foot toes against the wall and the right foot toes against left heel, the 2nd/3rd toe in line with left heel. The elbows should be at 90 degrees and in line with the rib cage. Once in position, lift the left foot off the ground to be weight-bearing on the right foot. Plantarflex right ankle. Switch legs for set-up and then repeat with left ankle.

Functional Criteria:

Normal ankle plantarflexion is 40 degrees.

Additional Information:

Watch for weight to remain over 1st and 2nd rays through the plantarflexion - no rolling to outside of foot.

Possible Findings:

- FN: 40 degrees of ankle plantarflexion in tandem stance, no pain
 - » Consider ankle plantarflexion normal. Proceed to Active Seated Ankle Inversion and Eversion Test.
- FP: 40 degrees of ankle plantarflexion in tandem stance, pain
 - » Proceed to the Passive Prone Plantarflexion Test.
- DN: Less than 40 degrees of ankle plantarflexion in tandem stance, no pain
 - » Proceed to the Passive Prone Plantarflexion Test.
- DP: Less than 40 degrees of ankle plantarflexion in tandem stance, pain
 - » Proceed to the Passive Prone Plantarflexion Test.



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Passive Prone Plantarflexion Test:

To test passive ankle plantarflexion in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Plantarflexion Test:

Instruct the patient to lie prone with foot off the edge of the table. Measure the total passive plantarflexion of the ankle.

Functional Criteria:

Normal open chain ankle plantarflexion is 40 degrees.

Additional Information:

Before taking the measurement, make sure the ankle starts in a neutral position.

Possible Findings:

- FN: 40 degrees of passive ankle plantarflexion, no pain
 - » Indicates an ankle plantarflexion stability/motor control dysfunction. Proceed to the Active Seated Ankle Inversion/Eversion Test.
- FP: 40 degrees of passive ankle plantarflexion, pain
 - » Indicates pain in the ankle with ankle plantarflexion - perform local biomechanical exam. Proceed to the Active Seated Ankle Inversion/Eversion Test.
- DN: Less than 40 degrees of ankle plantarflexion, no pain
 - » Indicates an ankle plantarflexion mobility dysfunction - perform local biomechanical exam. Proceed to the Active Seated Ankle Inversion/Eversion Test.
- DP: Less than 40 degrees of ankle plantarflexion, pain
 - » Indicates a painful ankle plantarflexion mobility dysfunction - perform local biomechanical exam. Proceed to the Active Seated Ankle Inversion/Eversion Test.



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Active Seated Ankle Inversion/Eversion Test:

To test active ankle inversion (supination) and eversion (pronation).

How to Perform the Active Seated Ankle Inversion/Eversion Test:

Instruct the patient to sit on a chair with the knees pelvic-width apart, feet flat on the ground, toes pointing forward and the hands on hips. Then, instruct the patient to invert and evert the feet. Have the person move back and forth between inversion and eversion.

Functional Criteria:

Normal ankle inversion/eversion is the ability to complete the motions. This is a quality test, not quantity - can they do it, or can't they.

Additional Information:

Because of ankle-joint limitations, many people create this motion at the hips. The person should be able to do this without moving the hips or knees.

Possible Findings:

- FN: Ability to actively invert/evert the ankle, no pain
 - » If the finding is FN for inversion and eversion, and there have been no other dysfunctions or pains, consider the finding a proprioceptive deficit.
- FP: Ability to actively invert/evert the ankle, pain
 - » Proceed to the Passive Seated Ankle Inversion/Eversion Test.
- DN: Inability to actively invert and/or evert the ankle, no pain
 - » Proceed to the Passive Seated Ankle Inversion/Eversion Test.
- DP: Inability to invert and/or evert the ankle, pain
 - » Proceed to the Passive Seated Ankle Inversion/Eversion Test.



SINGLE-LEG STANCE BREAKOUTS



Test Objective for the Passive Ankle Inversion/Eversion Test:

To test passive inversion (supination) and eversion (pronation) to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Ankle Inversion/Eversion Test:

Instruct the patient to lie supine on table. Stabilize the patient's lower limb and passively invert and evert their feet.

Functional Criteria:

Normal ankle inversion/eversion is the ability to complete the motions. This is a quality test, not quantity - can you do it, or can't you do it.

Additional Information:

You only need to test the directions that were dysfunctional or painful in the Active Seated Inversion/Eversion Test.

Possible Findings:

- FN: Ability to passively invert and/or evert the ankle, no pain
 - » Indicates an ankle inversion and/or eversion (depending on the motion you are testing) stability/motor control dysfunction.
- FP: Ability to passively invert and/or evert the ankle, pain
 - » Indicates pain with ankle inversion and/or eversion (depending on the motion you are testing) - perform local biomechanical exam.
- DN: Inability to passively invert and/or evert the ankle, no pain
 - » Indicates an ankle inversion and/or eversion (depending on the motion you are testing) mobility dysfunction - perform local biomechanical exam.
- DP: An inability to passively invert and/or evert the ankle, pain
 - » Indicates a painful ankle inversion and/or eversion (depending on the motion you are testing) mobility dysfunction - perform local biomechanical exam.

